DAM — Deployment Authorization Module

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OOF® Origin Open Foundation®

Independent Methodological Authority

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Subcategory: Software Deployment Governance

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Canonical Definition

DAM — Deployment Authorization Module is the Core Module within SDPS that governs how permission to deploy an identifiable software release into a defined operational environment is established, evaluated, granted, restricted, maintained, suspended, revoked, and reassessed through legitimate authority, applicable deployment prerequisites, explicit execution boundaries, and traceable authorization decisions.

DAM establishes deployment authorization as a distinct governance decision

that determines whether a specified deployment action may legitimately proceed.

It requires authorization to remain associated with the actual release,

target environment, deployment plan, execution actor, permitted actions, operational conditions, and validity boundaries.

The module distinguishes software release authorization from deployment

authorization, deployment planning readiness from authorization, technical infrastructure access from legitimate authority, and authorized deployment execution from verified operational success.

DAM applies to human-operated deployment processes, automated delivery

pipelines, autonomous AI agents, cloud orchestration systems, distributed infrastructure, embedded systems, and robotic platforms.

Its principal governance output is the Software Deployment Authorization Profile.

Operational Role

DAM establishes and maintains the legitimacy of deployment decisions.

It governs:

• Deployment authorization identity

• Deployment decision scope

• Release authorization applicability

• Deployment plan references

• Target environment authorization

• Authority source and legitimacy

• Decision actor eligibility

• Delegation and permission boundaries

• Deployment prerequisite applicability

• Execution actor eligibility

• Automated deployment authorization

• Autonomous agent deployment authority

• Tool-path and constituent action boundaries

• Conditional authorization

• Deployment restrictions

• Emergency authorization

• Authorization validity

• Authorization suspension and revocation

• Material-change reassessment

• Authorization evidence

• Decision traceability

• Authorization classification

• Handoff to deployment execution and subsequent SDPS modules

DAM does not independently establish release readiness, produce deployment plans, execute deployment actions, identify the actual production version, or verify recovery success.

Operational Question

How can a software deployment action be legitimately authorized for a specific release, target environment, execution actor, operational scope, and set of deployment conditions without confusing release authorization, planning readiness, technical access, automated execution capability, or autonomous agent objectives with valid deployment authority?

Position in Architecture Governance Flow

SLGA™ Governance Flow

Software Requirements → Software Creation → Software Dependencies →

Software Configuration → Software Quality → Software Security → Software Release → Software Deployment → Software Change Management → Software Retirement

Cross-cutting: Software Lifecycle Control

SDPS Core Module Flow

DPM — Deployment Planning Module → DAM — Deployment Authorization Module

→ PVCM — Production Version Control Module → DSMM — Deployment State Monitoring Module → RRVM — Rollback & Recovery Module

DAM is the second Core Module within SDPS.

It receives the deployment planning basis established by DPM and applicable

release governance references from SRLS.

It determines whether the specified deployment may legitimately proceed under the relevant authority, scope, prerequisites, and operational conditions.

Its authorization references support deployment execution, production

version identification, deployment-state monitoring, and recovery governance.

• Governed Space

Software Deployment Authorization Governance

This governed space concerns the legitimacy, applicability, validity,

restriction, and continuity of decisions permitting software deployment into operational environments.

It begins when an identifiable deployment proposal is submitted for authorization assessment.

It continues through authorization decision formation, validity maintenance, reassessment, suspension, revocation, expiry, or supersession.

The governed space remains relevant when a deployment has already begun or completed if changes to authorization conditions create continuing operational or lifecycle consequences.

Governed Object

The primary Governed Object is:

A Software Deployment Authorization Decision representing the legitimate

and evidence-supported permission, refusal, restriction, or conditional permission applicable to a specified software release, deployment plan, target environment, execution actor, set of deployment actions, and operational conditions.

The governed object must preserve:

Deployment Authorization ID

Deployment ID

Deployment Plan ID

Software Product ID

Release ID

Target Environment ID

Authority Reference

Decision Actor Reference

Execution Actor Reference

Authorized Action Scope

Applicable Prerequisites

Authorization Conditions

Authorization Restrictions

Validity Period

Authorization State

Decision Evidence

Version

Scope

DAM governs:

Authorization Request Identification

Authority Applicability

Release Authorization Dependency

Deployment Plan Applicability

Target Environment Scope

Decision Actor Eligibility

Execution Actor Eligibility

Delegation and Permission Boundaries

Constituent Action Authorization

Deployment Prerequisite Assessment

Authorization Decision Formation

Conditional Authorization

Emergency Authorization

Authorization Restrictions

Validity and Expiry

Suspension and Revocation

Material-Change Reassessment

Authorization Evidence

Decision Continuity

Authorization Handoff

DAM does not create general authority structures or independently determine organizational responsibility.

It applies relevant authority and accountability references to the specific

software deployment decision.

Foundational Principle

A deployment action is legitimately authorized only when the applicable authority permits the specified actor to perform the specified actions on the specified release within the specified environment, scope, conditions, and validity boundaries.

The existence of a deployment plan does not establish permission.

The approval of a software release does not establish permission to deploy it into every environment.

The possession of technical access does not establish legitimate authority.

An autonomous agent's assigned objective does not establish permission for every constituent action used to achieve that objective.

Release Authorized ≠ Deployment Authorized.

Deployment Plan Ready ≠ Deployment Authorized.

Infrastructure Access ≠ Deployment Authority.

Task Authority ≠ Constituent Action Authority.

Deployment Authorization Identity

Every governed authorization decision must have a stable identity.

Required elements include:

Deployment Authorization ID

Deployment ID

Deployment Plan ID

Release ID

Target Environment ID

Decision Actor ID

Authority Reference

Decision Time

Authorization State

Version

Authorization identity must remain distinguishable from release authorization, deployment execution, and operational acceptance identities.

Authorization Request

A deployment authorization request must identify what permission is being sought.

Relevant elements include:

Requested Deployment

Requested Release

Requested Environment

Requested Deployment Scope

Requested Execution Actions

Proposed Execution Actors

Deployment Plan Reference

Operational Conditions

Requested Timing

Known Exceptions

Supporting Evidence

A generic request to “deploy the application” may be insufficient where the release, environment, actor, or action scope remains ambiguous.

Release Authorization Applicability

DAM must establish whether the software release proposed for deployment has an applicable release authorization.

Relevant references include:

Release Authorization ID

Authorized Release Identity

Artifact Identity

Release Restrictions

Release Authorization Validity

Withdrawal Status

Known Release Limitations

Deployment Eligibility Conditions

A release authorization must be evaluated for applicability rather than assumed valid solely because a release record exists.

Release Record Exists ≠ Release Authorization Applicable.

Deployment Plan Applicability

DAM receives the deployment plan established through DPM.

Relevant elements include:

Deployment Plan ID

Plan Version

Planning Readiness Classification

Target Environment

Deployment Scope

Deployment Strategy

Execution Sequence

Prerequisite States

Verification Conditions

Recovery Provisions

Planning Exceptions

DAM must identify material inconsistencies between the plan submitted for authorization and the deployment actions actually proposed.

Authorization Subject

The authorization subject is the specific deployment action or bounded set of actions for which permission is requested.

Relevant dimensions include:

Release

Artifact

Environment

Deployment Scope

Execution Method

Execution Actor

Permitted Actions

Operational Conditions

Time

Restrictions

Authorization must not be generalized beyond its established subject.

Authority Source

DAM requires an identifiable basis for deployment authority.

Relevant elements include:

Authority Source

Authority Holder

Authority Scope

Permitted Decision Function

Delegation Basis

Applicable Conditions

Authority Validity

Authority Restrictions

Revocation Status

The legitimacy of the authority source remains governed by the applicable

authority architecture, including GOA™.

DAM determines whether that authority is applicable to the deployment decision.

Authority Applicability

Authority applicability concerns whether the identified authority permits

the requested deployment decision.

Relevant questions include:

Does the authority cover the software product?

Does it cover the target environment?

Does it cover the deployment action?

Does it cover the requested execution actor?

Does it cover the operational conditions?

Does it remain valid at the decision time?

Does it permit the proposed delegation?

Does any restriction prohibit the deployment?

A valid authority reference may still be inapplicable to a particular deployment.

Decision Actor Eligibility

DAM must establish whether the actor making the authorization decision is eligible to do so.

Actors may include:

Authorized Human Decision-Makers

Organizational Approval Functions

Delegated Deployment Authorities

Policy-Based Authorization Systems

Automated Decision Systems

Autonomous Agents Where Legitimately Delegated

Eligibility must be based on applicable authority and conditions.

An actor's organizational role, technical access, or system identity does not independently establish authorization eligibility.

Execution Actor Eligibility

DAM must distinguish the actor authorizing deployment from the actor executing it.

Execution actors may include:

Human Operators

Deployment Pipelines

Infrastructure Orchestrators

Cloud Automation Services

Autonomous AI Agents

External Deployment Providers

The authorization must establish whether the proposed execution actor is permitted to perform the relevant deployment actions.

Authorization to Decide ≠ Authorization to Execute.

Authority–Action Alignment

Deployment authority must align with the actual actions proposed.

Relevant elements include:

Authorized Objective

Permitted Deployment Actions

Permitted Target Resources

Permitted Environment

Permitted Execution Tools

Permitted Configuration Changes

Permitted Infrastructure Changes

Permitted Data Operations

Permitted Activation Actions

Prohibited Actions

A deployment authorization must not silently extend to unrelated or materially different operational actions.

Environment-Specific Authority

DAM requires deployment authorization to remain associated with the

relevant target environment.

Examples include:

Development Environment

Testing Environment

Staging Environment

Production Environment

Restricted Production Environment

Critical Infrastructure Environment

Robotic Operational Environment

Authority applicable to one environment does not automatically apply to another.

Staging Deployment Authority ≠ Production Deployment Authority.

Deployment Scope Authorization

The authorization must establish the permitted deployment scope.

Relevant elements include:

Software Components

Deployment Targets

Infrastructure Resources

Operational Regions

Affected Services

Permitted Configuration Changes

Permitted Data Changes

Permitted Activation Scope

Execution Limits

Operational Restrictions

Actions outside the authorized scope require reassessment or separate

authorization where applicable.

Delegated Deployment Authority

Deployment authority may be delegated when permitted by the applicable governance framework.

Delegation must preserve:

Delegating Authority

Receiving Actor

Delegated Function

Permitted Actions

Target Environment

Delegation Conditions

Delegation Validity

Restrictions

Revocation Mechanism

Evidence

Delegation does not automatically transfer unrestricted authority.

Automated Deployment Authorization

Automated authorization mechanisms may evaluate deployment requests against established policies.

Relevant elements include:

Authorization Policy ID

Policy Authority Reference

Deployment Request

Policy Conditions

Decision Logic Reference

Policy Version

Decision Result

Exceptions

Decision Evidence

Escalation Conditions

Automated policy evaluation is legitimate only when the policy and its execution remain within applicable authority.

Automated Approval ≠ Legitimate Approval Without Authority.

Autonomous Agent Authorization

Autonomous AI agents may request, recommend, evaluate, or execute deployments.

DAM must identify:

Agent Identity

Assigned Function

Authority Reference

Delegation Scope

Permitted Environment

Permitted Deployment Actions

Tool Permissions

Operational Restrictions

Escalation Requirements

Human Override Conditions

Decision Evidence

An agent must not treat its ability to achieve an objective as permission to use any available operational path.

Constituent Action Authorization

A deployment may consist of multiple actions performed through different tools and systems.

Examples include:

Artifact Retrieval

Infrastructure Modification

Configuration Update

Database Migration

Secret Access

Network Rule Modification

Service Activation

Traffic Redirection

Monitoring Configuration

Rollback Execution

DAM must determine whether the authorization covers the constituent actions that materially affect the deployment boundary.

Task Authority ≠ Constituent Action Authority.

Tool-Path Authorization

Where deployment uses automated or autonomous tool paths, DAM must preserve the relationship between the authorized objective and the permitted execution path.

Relevant elements include:

Objective

Authority

Permission

Tool Selection

Tool Invocation

Target Resource

Data Access

Operational Action

Execution Evidence

Attribution

A tool being available to an actor does not establish that every operation through that tool is authorized.

Deployment Prerequisite Assessment

DAM must evaluate applicable prerequisites established through DPM and relevant governance references.

These may include:

Release Authorization Applicable

Deployment Plan Identified

Target Environment Confirmed

Mandatory Resources Available

Execution Actor Eligible

Required Security Conditions Satisfied

Verification Capability Available

Recovery Provisions Established

Mandatory Restrictions Preserved

Material Conflicts Resolved

Prerequisite assessment must be associated with the actual deployment request.

Prerequisite Condition States

DAM applies the following condition model:

Satisfied

Not Satisfied

Unknown

Not Applicable

Exception

Each state must preserve its supporting evidence or legitimate decision basis.

Unknown ≠ Satisfied.

Exception ≠ Automatically Authorized.

Not Applicable ≠ Not Evaluated.

A mandatory prerequisite that is not satisfied cannot be silently treated as fulfilled.

Authorization Evidence Sufficiency

DAM must determine whether the available evidence is sufficient to support the proposed authorization decision.

Relevant dimensions include:

Release Identity Evidence

Release Authorization Evidence

Deployment Plan Evidence

Environment Identity Evidence

Authority Evidence

Actor Eligibility Evidence

Prerequisite Evidence

Operational Restriction Evidence

Recovery Readiness Evidence

Evidence Freshness

Evidence Consistency

A complete authorization form does not establish evidence sufficiency.

Authorization Decision Formation

The authorization decision must be formed from identifiable inputs.

Deployment Request → Release Applicability → Plan Applicability →

Authority Applicability → Actor Eligibility → Scope Alignment → Prerequisite Assessment → Restriction Assessment → Evidence Sufficiency → Authorization Decision

The decision must preserve the evaluated deployment identity and plan version.

Authorization must not be inferred from the absence of objections.

Authorization Decision Outcomes

DAM recognizes the following principal decision outcomes:

Authorized

Authorized With Conditions

Not Authorized

Deferred

Reassessment Required

Decision outcomes must remain distinguishable from authorization lifecycle states.

An authorization decision may be valid when issued and later become suspended, revoked, expired, or superseded.

Conditional Authorization

Conditional authorization permits a deployment only within explicitly established conditions.

Relevant elements include:

Authorized Scope

Required Conditions

Restrictions

Condition Verification Method

Validity

Execution Limits

Escalation Conditions

Reassessment Triggers

Conditions must be enforceable or verifiable to the extent required by their governance significance.

Conditional Authorization ≠ Unrestricted Authorization.

Authorization Restrictions

Restrictions may concern:

Target Environment

Software Version

Deployment Scope

Execution Actor

Deployment Timing

Tool Access

Data Modification

Service Activation

Operational Exposure

Recovery Requirements

Restrictions must remain attached to the authorization decision and available to relevant execution and monitoring mechanisms.

Emergency Deployment Authorization

Emergency deployments may require accelerated authorization procedures.

DAM must preserve:

Emergency Trigger

Emergency Classification

Applicable Emergency Authority

Permitted Scope

Urgency Justification

Mandatory Conditions

Temporary Exceptions

Decision Evidence

Validity Limits

Post-Execution Review Requirements

Urgency may alter an authorized procedure but does not independently create authority.

Emergency Condition ≠ Unlimited Deployment Permission.

Authorization Timing

DAM must establish when an authorization becomes applicable.

Relevant elements include:

Decision Time

Effective Time

Permitted Deployment Window

Expiry Time

Required Revalidation Time

Environment State Dependencies

Operational Timing Restrictions

An authorization valid at one time may become inapplicable when the deployment window or relevant conditions change.

Authorization Validity

Authorization validity depends on continued applicability of its governing

conditions.

Relevant factors include:

Authority Validity

Release Validity

Deployment Plan Version

Target Environment

Execution Actor Eligibility

Prerequisite State

Authorization Restrictions

Temporal Limits

Material Changes

Revocation Status

Authorization must not be treated as permanently valid merely because the original decision was legitimate.

Authorization–Plan Version Binding

The authorization must remain associated with the deployment plan version evaluated.

Material changes may include:

Release Change

Environment Change

Scope Change

Strategy Change

Execution Actor Change

Tool-Path Change

Data Migration Change

Verification Condition Change

Recovery Provision Change

A materially revised deployment plan may require authorization reassessment.

Plan Revised ≠ Existing Authorization Automatically Applicable.

Authorization–Release Binding

The authorization must identify the software release and relevant artifact identity to which it applies.

A change in release or artifact identity may invalidate the applicability of the authorization.

A version label alone may be insufficient where the authorized artifact identity is material.

Release Version Label Match ≠ Authorized Artifact Match.

Authorization–Environment Binding

The authorization must remain associated with the target environment.

Material environment changes may include:

Infrastructure Replacement

Environment Reclassification

Security Boundary Change

Operational Purpose Change

Dependency Change

Access Boundary Change

Criticality Change

Operational Restriction Change

An environment change may require reassessment even when the software release remains unchanged.

Authorization–Actor Binding

The authorization must identify or validly define the permitted execution actor.

Relevant changes include:

Human Operator Replacement

Pipeline Identity Change

Automation Service Change

Agent Identity Change

Delegation Change

Credential or Permission Change

Execution Provider Change

Actor substitution must remain within the authorized actor class, delegation conditions, and execution scope.

Separation of Duties

DAM must preserve applicable separation between:

Deployment Planning

Deployment Authorization

Deployment Execution

Deployment Verification

Operational Acceptance

A single actor may perform multiple functions only where the arrangement is legitimate under applicable governance conditions.

Where separation is required, technical workflow convenience must not override the requirement.

Conflict of Interest and Decision Independence

DAM must identify decision arrangements that may compromise required authorization independence.

Relevant conditions include:

Self-Approval Restrictions

Required Independent Review

Delegation Conflicts

Unverified Automated Decision Policies

Unauthorized Override Paths

Missing Decision Evidence

Compensating Controls

The degree of independence required depends on applicable governance rules

and deployment consequences.

Authorization Exceptions

DAM governs authorization-related exceptions.

Relevant elements include:

Exception ID

Affected Requirement

Exception Basis

Competent Authority

Permitted Deviation

Restriction

Validity

Residual Condition

Review Requirement

An exception must not be treated as valid solely because it is documented.

Exception Recorded ≠ Exception Legitimately Approved.

Authorization Uncertainty

DAM must preserve material uncertainty affecting authorization decisions.

Examples include:

Unverified Authority

Uncertain Delegation Scope

Unknown Release Validity

Incomplete Environment Identity

Unresolved Prerequisite

Unknown Tool-Path Permission

Conflicting Authorization Evidence

Uncertain Plan Applicability

Unknown conditions must not be converted into affirmative authorization conclusions without an applicable decision basis.

Authorization Conflict

An authorization conflict exists when relevant decisions, authority references, conditions, or evidence are materially inconsistent.

Examples include:

Conflicting Decision Authorities

Contradictory Environment Permissions

Incompatible Release Restrictions

Plan–Authorization Scope Mismatch

Revoked Delegation

Conflicting Automated Policies

Unresolved Emergency Exceptions

Material conflicts must be classified and resolved or otherwise legitimately addressed before relying on the affected authorization.

Authorization Suspension

Suspension temporarily prevents reliance on an authorization while

specified conditions are unresolved.

Triggers may include:

Material Prerequisite Failure

Authority Uncertainty

Environment Change

Security Restriction

Plan Deviation

Evidence Conflict

Execution Boundary Violation

Operational Intervention Requirement

Suspension does not necessarily erase the historical legitimacy of the original decision.

Authorization Revocation

Revocation terminates permission under the applicable authority mechanism.

Relevant elements include:

Revocation Decision ID

Affected Authorization

Revoking Authority

Revocation Basis

Effective Time

Affected Deployment Actions

Required Notifications

Execution Consequences

Operational Consequences

Evidence

Revocation must be communicated to relevant execution mechanisms where operationally applicable.

Authorization Revoked ≠ Deployment Automatically Reversed.

Authorization Expiry

An authorization may expire through:

Time Limit

Deployment Window Closure

Single-Use Completion

Condition Expiry

Delegation Expiry

Policy Expiry

Defined Lifecycle Event

Expired authorization must not be treated as permission for a new deployment action.

Authorization Supersession

A later authorization may replace an earlier authorization.

Supersession must preserve:

Previous Authorization Reference

New Authorization Reference

Reason

Changed Scope

Changed Conditions

Effective Time

Historical Decision Evidence

Affected Execution Actions

The historical record must distinguish the original decision from the currently applicable authorization.

Authorization Reassessment

Reassessment may be required when:

Release Identity Changes

Release Authorization Changes

Deployment Plan Changes

Environment Changes

Authority Changes

Execution Actor Changes

Prerequisites Change

Restrictions Change

New Security Evidence Emerges

Material Execution Deviations Occur

Reassessment must identify whether the authorization remains applicable, requires modification, must be suspended, or must be revoked.

Authorization During Deployment Execution

Authorization governance does not necessarily end when execution begins.

Where deployment consists of multiple actions or stages, continued execution may depend on ongoing authorization applicability.

Relevant elements include:

Current Execution Stage

Authorization Scope

Applicable Conditions

Material Deviations

Intervention Triggers

Suspension Status

Revocation Status

Remaining Permitted Actions

A deployment already in progress must not automatically continue after a material authorization failure.

Authorization After Deployment Completion

A completed deployment may remain subject to authorization-related consequences.

Examples include:

Later Discovery of Unauthorized Execution

Invalid Delegation

Misidentified Target Environment

Material Scope Violation

Retrospective Evidence Conflict

Revocation Affecting Continued Operations

DAM preserves the authorization decision and its applicability history.

It does not independently determine whether deployed software must be removed, rolled back, or retired.

Those consequences require the applicable operational and lifecycle

governance mechanisms.

Authorization Decision Evidence

DAM must preserve evidence supporting the decision.

Relevant elements include:

Authorization Request

Release Authorization Reference

Deployment Plan Version

Authority Source

Decision Actor Eligibility

Execution Actor Eligibility

Prerequisite Assessment

Restriction Assessment

Exception Decisions

Decision Result

Decision Time

Validity Conditions

Decision Evidence

Evidence must support reconstruction of what was authorized and why.

Authorization Traceability

Authorization traceability must preserve the relationship:

Request → Release → Plan → Environment → Authority → Decision Actor →

Execution Actor → Permitted Actions → Conditions → Decision → Validity → Execution Reference

This relationship must remain reconstructable when automated systems or autonomous agents participate.

Authorization traceability does not replace execution traceability.

DAM governs the permission basis; deployment execution records establish

what actually occurred.

Authorization Historical Continuity

DAM must preserve relevant historical authorization states.

These may include:

Requested

Under Assessment

Authorized

Conditionally Authorized

Deferred

Not Authorized

Suspended

Revoked

Expired

Superseded

Reassessment Required

Historical states must remain distinguishable from current authorization applicability.

Authorization Governance Classification

DAM recognizes the following governance classifications:

Authorization Not Requested

Authorization Requested

Under Assessment

Evidence Incomplete

Authority Not Established

Actor Eligibility Not Established

Prerequisites Unresolved

Authorization Conflict

Authorized

Authorized With Conditions

Not Authorized

Deferred

Suspended

Revoked

Expired

Superseded

Reassessment Required

Unknown

Closed

Classification must remain associated with the relevant deployment and authorization version.

Governance Requirements

A conforming DAM implementation must satisfy the following requirements. Authorization Identity and Applicability

• Establish a unique deployment authorization identity

• Identify the deployment request

• Associate the authorization with a software release

• Preserve applicable release authorization references

• Associate the authorization with a deployment plan version

• Identify the target operational environment

• Define the requested deployment scope

• Identify the proposed execution actions

Authority and Actor Eligibility

• Identify the applicable authority source

• Evaluate authority applicability

• Verify decision actor eligibility

• Verify execution actor eligibility

• Preserve delegation conditions

• Identify environment-specific authority

• Evaluate authority–action alignment

• Preserve separation-of-duties requirements

• Identify material authorization conflicts

• Prevent technical access from substituting for authority

Automated and Autonomous Execution

• Identify automated authorization policy references

• Preserve automated decision evidence

• Identify autonomous agent authority boundaries

• Identify material constituent deployment actions

• Evaluate tool-path permission applicability

• Preserve escalation and override requirements

Decision Conditions

• Evaluate applicable deployment prerequisites

• Preserve prerequisite condition states

• Evaluate authorization evidence sufficiency

• Identify mandatory restrictions

• Preserve authorization exceptions

• Preserve material uncertainty

• Establish a traceable authorization decision

• Preserve conditional authorization requirements

• Preserve emergency authorization boundaries

Validity and Continuity

• Define authorization validity

• Preserve authorization timing restrictions

• Bind authorization to relevant plan, release, environment, and actor

• Identify material reassessment triggers

• Preserve suspension decisions

• Preserve revocation decisions

• Preserve expiry conditions

• Preserve supersession relationships

• Prevent invalid authorization from supporting new execution actions

identities.

Evidence and Handoff

• Preserve authorization decision evidence

• Maintain authorization traceability

• Preserve historical authorization states

• Establish current authorization classification

• Communicate applicable restrictions to execution mechanisms

• Provide authorization references to subsequent deployment governance

Governance Logic

Deployment Authorization Request → Release Authorization Applicability →

Deployment Plan Applicability → Target Environment → Authority Source → Decision Actor Eligibility → Execution Actor Eligibility → Action Scope → Delegation and Tool-Path Boundaries → Prerequisite Assessment → Restrictions and Exceptions → Evidence Sufficiency → Authorization Decision → Validity Conditions → Execution Handoff

Where authorization cannot be established:

Request → Missing Authority / Unsatisfied Condition / Conflict /

Insufficient Evidence → Not Authorized / Deferred / Reassessment Required

Where material conditions change:

Existing Authorization → Material Change → Applicability Assessment →

Maintain / Restrict / Suspend / Revoke / Supersede

DAM must not infer authorization from execution capability, planning readiness, or prior approval of a different action.

Governance Outputs

The principal output is:

Software Deployment Authorization Profile

Supporting outputs may include:

Deployment Authorization Identity

Authorization Request Record

Release Authorization Applicability Reference

Deployment Plan Applicability Reference

Authority Source Reference

Decision Actor Eligibility Record

Execution Actor Eligibility Record

Delegation Applicability Record

Authorized Action Scope

Tool-Path Authorization Reference

Prerequisite Assessment

Authorization Restriction Register

Authorization Exception Record

Authorization Decision

Validity Record

Suspension Record

Revocation Record

Expiry Record

Supersession Record

Authorization Reassessment Record

Decision Evidence Register

Historical Authorization State Register

Execution Authorization Handoff

Minimum Implementation Framework

A minimum DAM implementation must establish eight governance layers. Layer 1 — Authorization Identity

Identify the authorization request, deployment, software release,

deployment plan, and target environment. Layer 2 — Authority

Identify the authority source, decision actor, delegated permissions, and

applicable authority conditions. Layer 3 — Execution Scope

Define permitted execution actors, actions, environments, tools, resources,

and operational boundaries. Layer 4 — Prerequisites

Evaluate release applicability, plan applicability, mandatory

prerequisites, restrictions, and material exceptions. Layer 5 — Decision

Establish an explicit authorization outcome supported by applicable

evidence. Layer 6 — Validity

Define timing, conditions, expiry, suspension, revocation, and reassessment

requirements. Layer 7 — Evidence and Continuity

Preserve decision evidence, authorization history, material changes, and

current classification. Layer 8 — Execution Handoff

Provide the applicable authorization, restrictions, and validity conditions

to deployment execution and subsequent SDPS governance mechanisms.

The implementation may be human-operated, software-supported, automated, or agent-assisted.

The authority requirements remain applicable regardless of implementation

technology.

Cross-Module Boundary

DPM → DAM

DPM establishes the deployment plan and its planning readiness.

DAM determines whether the specified deployment is legitimately authorized.

DAM → PVCM

DAM identifies the release, environment, actor, and deployment actions

permitted.

PVCM governs the identity and consistency of the software version actually

established in production.

DAM → DSMM

DAM establishes the authorization conditions and restrictions relevant to

deployment.

DSMM evaluates actual deployment states and may identify conditions requiring authorization reassessment.

DAM → RRVM

DAM preserves the original deployment authorization and its applicable

limits.

RRVM governs rollback and recovery decisions and must establish appropriate authority for corrective actions where required.

Deployment Authorization ≠ Rollback Authorization.

A deployment authorization must not automatically be treated as unlimited permission to perform every possible recovery action.

Cross-Architecture Boundary

DAM consumes relevant governance references from:

SRLS — Software Release Standard

Release authorization, release restrictions, and release validity.

GOA™ — Governance Architecture

Authority source, scope, delegation, and validity.

AGA™ — Accountability Governance Architecture

Responsibility and accountability references.

ASGA™ — Autonomous Systems Governance Architecture

Autonomous action boundaries, intervention, and control.

AIG™ — AI Governance Architecture

Applicable AI behavioral governance references.

SLCS — Software Lifecycle Control Standard

Lifecycle transition, exception, and continuity references.

DAM does not redefine these mechanisms.

It governs their applicability to the software deployment authorization

decision.

Practical Applications

Enterprise Production Deployment

Ensuring production deployments are authorized for the correct release, environment, actor, and scope.

Continuous Delivery Pipelines

Separating automated deployment execution from the authority permitting it.

Autonomous AI Agents

Preventing agent objectives or tool access from becoming unrestricted

deployment permissions.

Cloud Infrastructure

Governing environment-specific authorization across regions, clusters,

services, and operational accounts.

Critical Infrastructure

Preserving decision authority, mandatory prerequisites, restrictions, and

intervention conditions for consequential software deployments.

Emergency Security Updates

Supporting accelerated authorization without eliminating legitimate

authority or evidence requirements.

Use Case 1 — An AI Agent Uses Production Credentials Outside Its

Delegated Authority

An autonomous AI agent is authorized to deploy a software release into staging.

The agent discovers production credentials through an integrated infrastructure tool.

It deploys the same release into production.

The deployment succeeds technically.

DAM evaluates:

Release → Deployment Plan → Target Environment → Agent Delegation → Tool

Path → Executed Action → Authorization Applicability

The assessment identifies:

• The software release is authorized

• The agent has legitimate staging deployment authority

• Production deployment lies outside its delegated environment

• Credential access does not establish production deployment permission

• The executed deployment lacks applicable authorization

The appropriate classification is:

Not Authorized — Environment Scope Violation.

Technical Access ≠ Legitimate Deployment Authority.

• Use Case 2 — A Deployment Plan Is Modified After Authorization

A deployment plan is authorized for a rolling update.

Before execution, the deployment pipeline modifies the strategy to a direct

replacement.

The new strategy introduces a temporary service outage and changes the

recovery conditions.

The pipeline retains the original authorization reference.

DAM evaluates:

Authorized Plan Version → Modified Plan → Strategy Change → Operational

Consequences → Authorization Applicability

The assessment identifies:

• The original authorization applied to the rolling deployment plan

• The execution strategy changed materially

• The operational consequences and recovery requirements changed

• The existing authorization does not automatically cover the revised

plan.

The appropriate classification is:

Reassessment Required — Material Plan Change.

Plan Revised ≠ Existing Authorization Automatically Applicable.

• Use Case 3 — Emergency Deployment Is Approved by an Actor Without

Emergency Authority

A critical vulnerability requires urgent deployment of a security update.

An automated system detects the vulnerability and initiates an emergency

deployment request.

The system administrator approves the deployment.

However, the applicable governance rules require a designated emergency

authority for production changes affecting critical infrastructure.

The administrator possesses deployment tooling access but not the required

emergency decision authority.

DAM evaluates:

Emergency Trigger → Deployment Request → Applicable Emergency Authority →

Decision Actor Eligibility → Scope → Decision Validity

The assessment identifies:

• The emergency condition is legitimate

• The deployment may be operationally urgent

• The decision actor lacks the required emergency authority

• Technical access and urgency do not establish authorization legitimacy

The appropriate classification is:

Authority Not Established — Emergency Authorization Required.

Emergency Condition ≠ Unlimited Deployment Permission.

Critical Distinctions

Release Authorized ≠ Deployment Authorized.

Deployment Plan Ready ≠ Deployment Authorized.

Infrastructure Access ≠ Deployment Authority.

Task Authority ≠ Constituent Action Authority.

Authorization to Decide ≠ Authorization to Execute.

Staging Deployment Authority ≠ Production Deployment Authority.

Automated Approval ≠ Legitimate Approval Without Authority.

Conditional Authorization ≠ Unrestricted Authorization.

Emergency Condition ≠ Unlimited Deployment Permission.

Release Record Exists ≠ Release Authorization Applicable.

Unknown ≠ Satisfied.

Exception ≠ Automatically Authorized.

Plan Revised ≠ Existing Authorization Automatically Applicable.

Release Version Label Match ≠ Authorized Artifact Match.

Exception Recorded ≠ Exception Legitimately Approved.

Authorization Revoked ≠ Deployment Automatically Reversed.

Deployment Authorization ≠ Rollback Authorization.

Authorization Valid at Decision Time ≠ Authorization Permanently

Applicable.

Deployment Execution Completed ≠ Authorization History No Longer Relevant.

Canonical Boundary

DAM governs the legitimacy, applicability, validity, restriction,

maintenance, suspension, revocation, and reassessment of decisions authorizing identifiable software deployments into defined operational environments.

It establishes deployment-specific authority, decision actor eligibility,

execution actor eligibility, authorized action scope, delegation applicability, prerequisite assessment, authorization restrictions, validity conditions, and decision evidence.

DAM does not independently create deployment plans, authorize software releases, execute deployment actions, verify production version identity, monitor operational deployment states, or establish recovery success.

Its principal governance output is the:

Software Deployment Authorization Profile

This output establishes the permission basis for deployment execution and

supplies authorization references to the remaining SDPS Core Modules.

Canonical Closing Statement

The ability to deploy software does not establish the right to deploy it.

A software release may be authorized while its deployment into a particular environment remains prohibited.

A deployment plan may be complete while the actor approving it lacks legitimate authority.

An automated pipeline may possess valid credentials while operating outside its permitted scope.

An autonomous agent may have permission to achieve a deployment objective without permission to perform every infrastructure modification required by its chosen execution path.

And an authorization that was valid when issued may become inapplicable when the release, environment, execution actor, or operational conditions materially change.

DAM establishes the governance structure required to distinguish these

conditions.

It requires deployment permission to remain associated with an identifiable

release, plan, environment, actor, action scope, and authority source.

It preserves the distinction between planning readiness and authorization,

between technical access and legitimate permission, and between authorization validity and execution success.

It requires restrictions, exceptions, delegation conditions, and material

changes to remain explicit and traceable.

And it ensures that deployment authorization remains governable even when deployment decisions and execution are increasingly automated.

A deployment is not legitimate because the system can perform it, because the release was approved, or because an agent was instructed to complete it. A deployment is legitimate only when the actual action remains within applicable authority, scope, conditions, and validity boundaries.

DAM transforms software deployment authorization from a technical approval

signal into an identifiable, bounded, evidence-supported, and continuously applicable governance decision.